

Homework — 1.3.2 Databases — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] MemberID in the Booking table is the foreign key (1). It links to MemberID, the primary key of the Member table (1).

2. [3] Sample answer:

```
SELECT FirstName, Surname
FROM Member
WHERE MembershipType = 'Junior'
ORDER BY Surname DESC;
```

1 mark: SELECT FirstName, Surname FROM Member; 1 mark: WHERE MembershipType = 'Junior'; 1 mark: ORDER BY Surname DESC (must include DESC).

3. [3] Sample answer:

```
SELECT Member.Surname, Booking.CourtNumber, Booking.BookingDate
FROM Member
JOIN Booking ON Member.MemberID = Booking.MemberID;
```

1 mark: SELECT lists Surname, CourtNumber and BookingDate; 1 mark: FROM/JOIN names both tables; 1 mark: correct join condition ON Member.MemberID = Booking.MemberID. (Accept table aliases, INNER JOIN, or an equivalent implicit join in a WHERE clause.)

4. (a) [2] A single order record would contain a repeating group — several ProductID / ProductName / Quantity values in one row — so values are not atomic (1). 1NF requires no repeating groups / every attribute to hold a single (atomic) value, with a primary key defined (1). (Accept: each product on an order must become its own row / separate order-line table.)

(b) [2] It is a **partial dependency** — a non-key attribute depends on only part of a composite primary key (1). It is removed by converting to **Second Normal Form (2NF)** (1).

5. [2] Durability: once a transaction has been committed, its changes are permanent — they are written to non-volatile storage (e.g. via a transaction log) (1). So if the server loses power after the commit, the payment is not lost — the committed change survives the failure/restart (1).

Part B (6 marks)

6. [2] Lossy compression (1). Valid reason (1), e.g. it produces a much smaller file, so the page loads faster / uses less bandwidth and storage, and the small loss of quality is not noticeable when viewing a photograph on screen.

7. [2] The Waterfall model is a sequential/linear series of stages (analysis, design, implementation, testing, maintenance) where each stage is completed before the next begins (1). Drawback (1, any one): inflexible — going back to change an earlier stage is difficult/costly; the customer only sees the product near the end; unsuitable when requirements are likely to change.

8. [2] 1 mark + 1 mark for a developed reason, e.g. an SSD has no moving parts, so data access is much faster (lower latency/read times) (1) and it is more resistant to damage from knocks/drops, uses less power (longer battery life) and is silent (1). (Two valid points, or one point well developed.)

Total: 14 + 6 = 20 marks